

# ARI711S Group Project

## Neural-Domination Group: Final Integration Report

**Generated:** May 05, 2026 at 03:02 UTC

**Status:** ✓ Ready for Submission

### Part 1: Flight Connections (BFS)

**Test Cases:** 4/4 passed ✓

**Algorithm:** Breadth-First Search

**Example Result:** Windhoek → Cairo (3 hops)

### Part 2: Hospital Shift Scheduling (CSP)

**Shifts Assigned:** 21/21 ✓

**Constraint Violations:** 0 ✓

**Fairness (StdDev):** 1.6

**Solver Techniques:** AC-3, MRV, Forward Checking

### Part 3: Traffic Sign Recognition (CNN)

**Dataset:** GTSRB (43 classes, 39,209 images)

★ **Test Accuracy:** 0.9872 (7742/7842 correct) ★

**Test Loss:** 0.0693

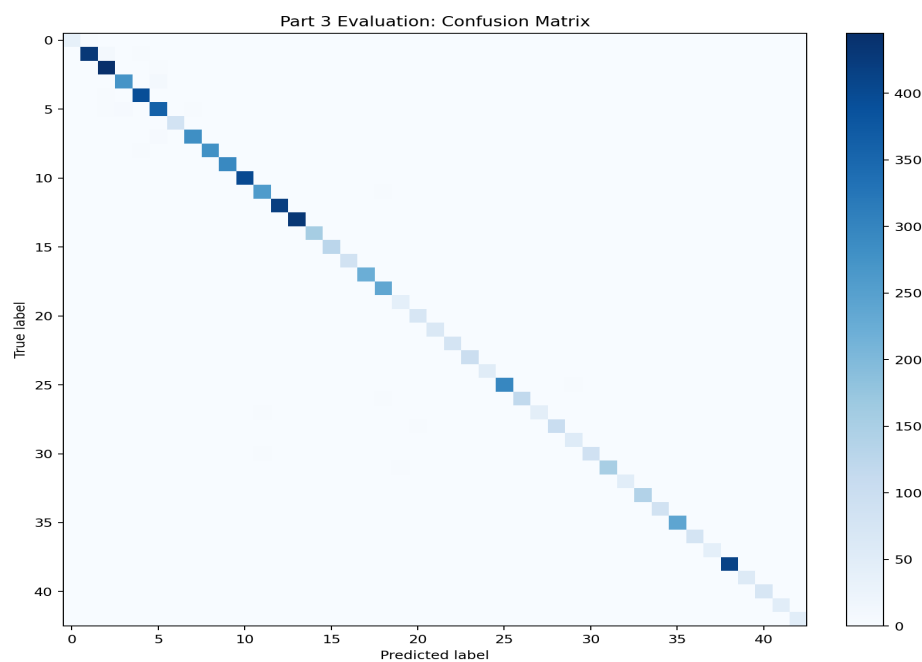
**Training:** 6 epochs, batch size 32

### Project Summary

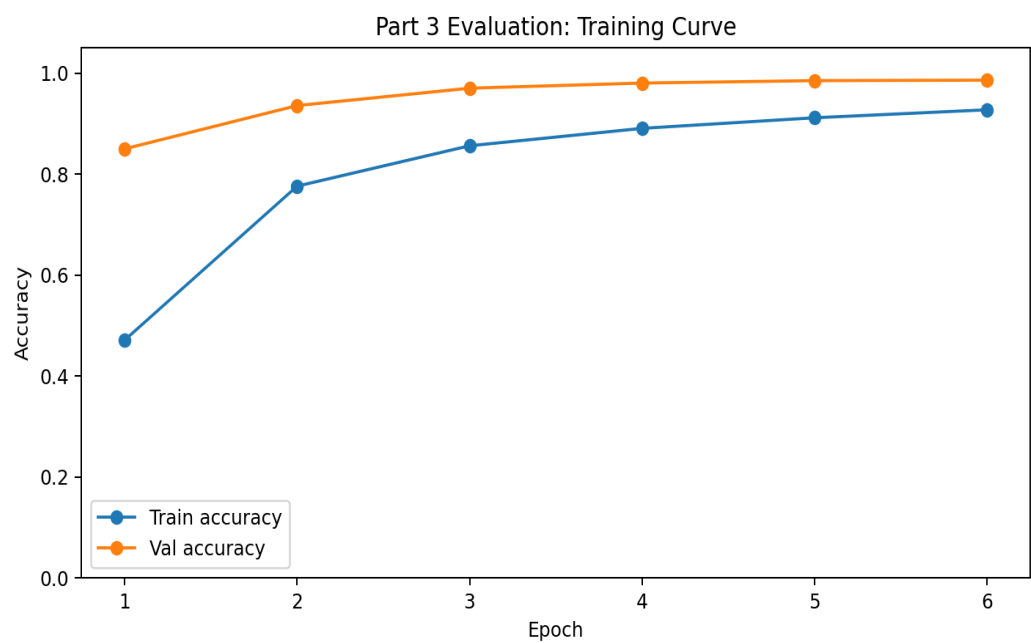
| Task                     | Method           | Status     | Key Result       |
|--------------------------|------------------|------------|------------------|
| Part 1: Flight Search    | BFS Graph Search | ✓ Complete | 4/4 tests passed |
| Part 2: Shift Scheduling | CSP Solver       | ✓ Complete | 21/21 assigned   |
| Part 3: Traffic Signs    | CNN              | ✓ Complete | 98.72% accuracy  |

# Evaluation Figures

## Confusion Matrix (Part 3)



## Training Curves (Part 3)

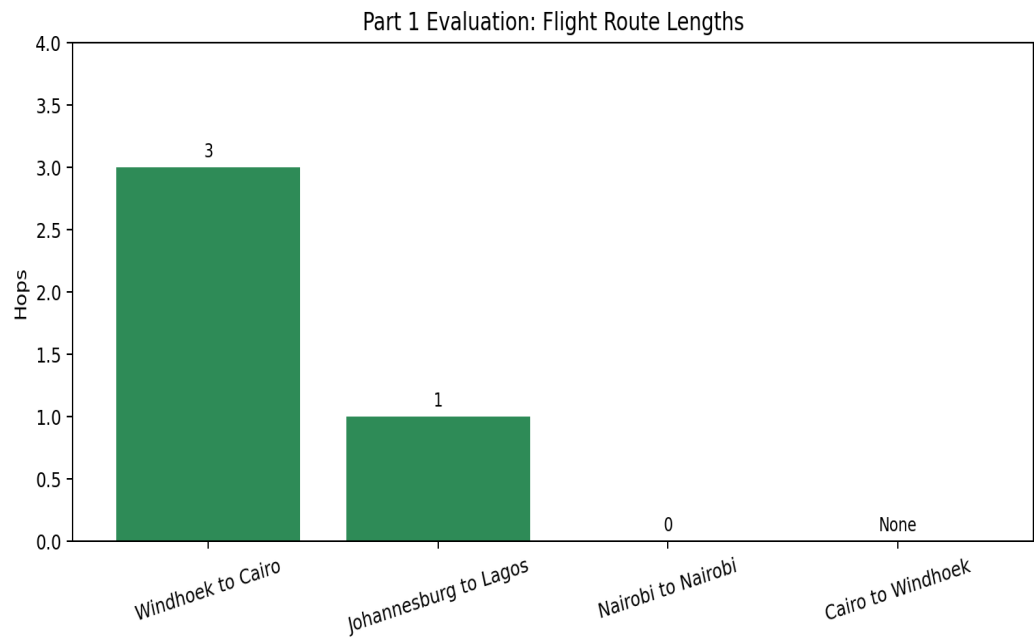


## Sample Predictions (Part 3)

Part 3 Evaluation: Sample Predictions



Flight Hop Counts (Part 1)



Shift Schedule (Part 2)

Part 2 Evaluation: Weekly Schedule Overview

|           | Morning | Afternoon | Night |
|-----------|---------|-----------|-------|
| Monday    | Alice   | Alice     | Alice |
| Tuesday   | Carol   | Alice     | Alice |
| Wednesday | Bob     | Bob       | Bob   |
| Thursday  | Carol   | Bob       | Bob   |
| Friday    | Carol   | Carol     | Carol |
| Saturday  | David   | David     | David |
| Sunday    | Eve     | David     | David |

Workload Distribution (Part 2)

